# Jodh Singh Nahal

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## Education

**University of British Columbia** *Vancouver, BC*
*Bachelor of Applied Science in Computer Engineering* *Expected May 2028*
- Presidential Scholars Award ($60,000), Dean's Honour List, available 4 or 8 months, CGPA: 84.7%.

## Technical Skills

**Languages**: Java, JavaScript, Python, C/C++, SQL
**Mobile & Connectivity**: Mobile-Compatible Web Apps, Bluetooth/BLE, REST, WebSocket, SQLite
**Tools**: Git, Flask, Node.js, Linux, Docker

## Experience

**Hardware Verification Engineering Intern** *Vancouver, BC*
*AMD* *Sept 2026 – Present*
- Built an AI-driven browser automation tool that auto-provisions tool access and software installs for onboarding.
- Contributing to verification of the IOHUB subsystem for next-generation AMD processor architectures.
- Verifying IOMMU L1 cache behavior using UVM in a constrained-random testbench.
- Expanded the constrained-random testbench to cover corner cases and improve functional coverage.

**Software Test / Embedded Development Co-op** *Burnaby, BC*
*Xantrex* *May 2024 – Aug 2025*
- Troubleshot CAN/BLE communication faults on embedded inverter and charger platforms using oscilloscopes, logic analyzers, PCAN-USB adapters, and BLE sniffers.
- Extended firmware support to NMEA 2000 and a proprietary UART battery communication protocol.
- Developed an object-oriented, multi-threaded, non-blocking C++ firmware architecture on Linux-based Raspberry Pi gateways to parse RV-C CAN frames from STM32 devices and publish structured messages over MQTT.
- Replaced a ∼60k-line Python implementation with the C++ architecture.
- Achieved 7× device capacity, 90% lower CPU usage, and ∼2 s lower latency after the C++ migration.

**Firmware / Embedded Systems Developer** *Vancouver, BC*
*UBC Rover* *Sept 2023 – July 2026*
- Automated firmware validation with GitHub Actions CI and Catch2 unit tests to improve regression coverage.
- Led a team of 3 engineers building a distributed embedded control system across ESP32 (FreeRTOS), STM32, Arduino, and TI LaunchPad MCUs over SPI/I2C.
- Aggregated inputs from switches, buttons, potentiometers, encoders, and joysticks and streamed data over UART.
- Developed a 16-channel telemetry system on a TI C2000 (LAUNCHXL-F28069M) using ePWM-triggered ADC interrupts, streaming GPIO/ADC data over SCI/UART.

**Undergraduate Research Assistant** *Vancouver, BC*
*UBC Electrical and Computer Engineering — Supervisor: Dr. Lutz Lampe* *May 2026 – Sept 2026*
- Optimizing power extraction of piezoelectric energy harvesting interface circuits (SEH, SSHI, SECE) through LTspice and Simulink/Simscape simulation.
- Researching battery-less platforms for the Internet of Underwater Things via simulation of piezoelectric interface circuits.
- Designing embedded power-management PCB hardware in KiCad integrating AC–DC rectifiers, transformers, and switching circuitry.

## Projects

**BlindVision** | *Flask, YOLOv8, OpenCV, Gemini API, JavaScript*
- Built a web application compatible with both mobile and desktop.
- Developed a full-stack IoT web app giving visually impaired users real-time AI-powered audio descriptions of live video and audio.

**Multimodal AI Messenger** | *Node.js, Express, WebSocket, MongoDB, Gemini API*
- Built a responsive frontend with media upload.